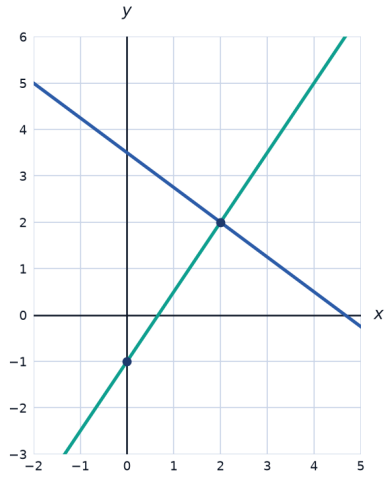
# Substitution into formulas

## Core Concepts: Substitution into formulas

**1** For the model $f(x) = 6x + 3$ in substitution into formulas, evaluate $f(2)$.




**a.** 15

**b.** 21

**c.** 12

**d.** 19

**2** Find the positive solutions to the quadratic equation $x^2 - 5x + 4 = 0$.

**a.** $x = 1$ and $x = 5$

**b.** $x = 1$ and $x = 4$

**c.** $x = -1$ and $x = -4$

**d.** $x = 5$ and $x = 4$

**3** A data set recorded for substitution into formulas consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

**a.** 18.5

**b.** 19.5

**c.** 20.5

**d.** 22.5

---

## Procedural Fluency & Algebraic Manipulation

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**4** For the model $f(x) = 6x + 3$ in substitution into formulas, evaluate $f(3)$.

**a.** 21

**b.** 27

**c.** 18

**d.** 25

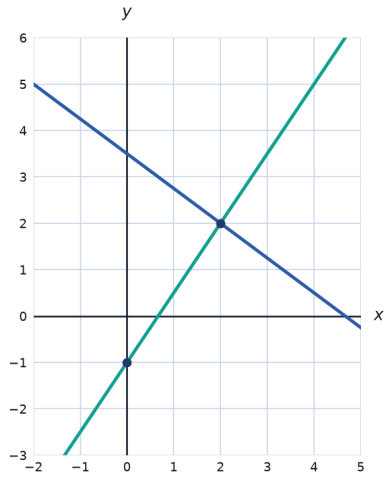
**5** Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.

**a.** $x = 2$ and $x = 6$

**b.** $x = 2$ and $x = 5$

**c.** $x = -2$ and $x = -5$

**d.** $x = 7$ and $x = 10$




**6** A data set recorded for substitution into formulas consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

**a.** 18.5

**b.** 19.5

**c.** 20.5

**d.** 22.5

## Graphical & Geometric Interpretation

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**7** For the model $f(x) = 6x + 3$ in substitution into formulas, evaluate $f(4)$.

**a.** 27

**b.** 33

**c.** 24

**d.** 31

**8** Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

**a.** $x = 3$ and $x = 7$

**b.** $x = 3$ and $x = 6$

**c.** $x = -3$ and $x = -6$

**d.** $x = 9$ and $x = 18$

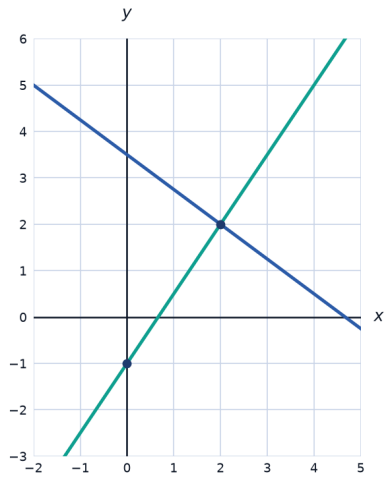
**9** A data set recorded for substitution into formulas consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

**a.** 18.5

**b.** 19.5

**c.** 20.5

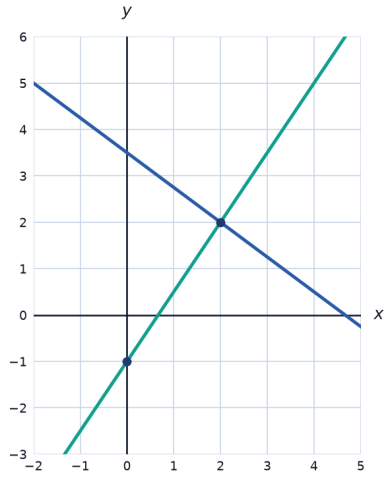
**d.** 22.5




## Exam-Style Problem Solving & Modeling

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**10** For the model $f(x) = 6x + 3$ in substitution into formulas, evaluate $f(5)$.




**a.** 33

**b.** 39

**c.** 30

**d.** 37

**11** Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

**a.** $x = 4$ and $x = 8$

**b.** $x = 4$ and $x = 7$

**c.** $x = -4$ and $x = -7$

**d.** $x = 11$ and $x = 28$

**12** A data set recorded for substitution into formulas consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

**a.** 18.5

**b.** 19.5

**c.** 20.5

**d.** 22.5

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